

greater cognitive control (Ofen et al., 2012). The absence of MTL activation aligns with the complex developmental trajectories and subregional differences characterizing early MTL development (Ofen et al., 2012). Planned ROI analyses targeting a priori (PFC, MTL) and empirical occipito-temporal peaks will further examine sub-threshold regional contributions to memory and vocabulary outcomes.

D27 Sandbox Series - Prior knowledge modulates the consolidation of nonnative tone learning during daytime naps in Mandarin-speaking older adults: the role of coupling strength

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Sleep-dependent memory consolidation supports language learning, but this may depend on learners' prior knowledge. For instance, Cantonese Tone 5 (T5)–T6 relies on a pitch contour cue (present in the Mandarin tone system, thus aligned with Mandarin speakers' prior L1 knowledge), whereas T3–T6 relies on a pitch height cue (absent in Mandarin, not aligned) (Qin et al., 2025). Liu (2025) used an overnight design and found that Mandarin-speaking older adults in the evening group (trained before sleep) showed improvement after sleep for Cantonese T5–T6 but not for T3–T6. This benefit was absent in the morning group (trained without subsequent sleep). However, Liu (2025) lacked time-of-day controls and micro-level sleep parameters measured by Electroencephalography (EEG). Older adults experienced reduced temporal precision in slow oscillation (SO)-spindle coupling (Helfrich et al., 2018), which may lead to greater individual variability and be critical for memory consolidation in this population. The present study uses a nap paradigm with EEG monitoring to examine (1) whether consolidation occurs during short naps in older adults, and (2) whether SO-spindle coupling plays a critical role in older adults' memory consolidation. Forty-two Mandarin-speaking older adults (55–78 years old) were pseudo-randomly assigned to a nap (n=19) or wake group (n=23). All participants were trained on Cantonese T5–T6 and T3–T6. Each contrast contained 4 words. This was followed by an identification test (160 trials). Then, during a 90-minute interval, the nap group took a daytime nap with EEG monitoring, while the wake group remained awake. EEG was recorded using the international 10-20 system with an actiCHamp Plus amplifier. Ten scalp electrodes (Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2) were used, along with vertical and horizontal electro-oculograms. The sampling rate was 1000 Hz, and the online reference electrode was Cz. Both groups were subsequently retested using the same identification test. Accuracy was analyzed with generalized mixed-effects models. We measured SO-spindle coupling strength using phase-amplitude coupling analysis. The relationship between coupling strength and accuracy was then examined. A three-way interaction among Session, Tone, and Group was found ($\beta=0.47$, $SE=0.17$, $z=2.71$, $p=.01$). Follow-up models showed that for T5–T6, the Session $\times$ Group interaction was significant ($\beta=-0.33$, $SE=0.14$, $z=-2.41$, $p=.02$). The decrease in accuracy was greater in the wake group than in the nap group (Wake: $\beta=0.60$, $SE=0.09$, $z=6.48$, $p<.001$; Nap: $\beta=0.27$, $SE=0.10$, $z=2.67$, $p=.01$). For T3–T6, only a main effect of Session emerged ($\beta=-0.13$, $SE=0.05$, $z=-2.48$, $p=.01$). No interaction was found ($\beta=0.12$, $SE=0.10$, $z=1.19$, $p=.23$). Furthermore,

coupling strength positively predicted accuracy changes in the nap group ($\beta=2.23$, $SE=1.04$, $z=2.14$, $p=.03$). Unlike overnight consolidation, which led to performance improvement (Liu, 2025), both contrasts declined at retest in both groups. However, T5–T6 showed a smaller decline than T3–T6 in the nap group, reflecting better consolidation. It demonstrates that prior knowledge modulates consolidation during a short nap in older adults. SO-spindle coupling played a critical role in consolidation. These results highlight how prior knowledge supports tonal consolidation in aging and suggest that SO-spindle coupling may serve as an early marker of age-related neural degradation in sleep-dependent consolidation.

D28 Sandbox Series - Procedural Memory Predicts Early Language Development: Domain-General and Production-Selective Effects with Preliminary fMRI Evidence

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Introduction. Procedural memory (PM), which underlies implicit sequence learning through repeated exposure, has been linked to grammatical development (Conti-Ramsden et al., 2015; Ullman, 2016). Yet it remains undercharacterized whether PM–language associations extend to vocabulary, which also involves systematic phonological-articulatory sequencing (Gupta & Dell, 1999), and whether they are stronger for production than comprehension given production's greater active sequencing demands, or vary across learning phases. Further, existing pediatric evidence for PM–language links is largely behavioral or based on school-aged children, leaving open how procedural learning is instantiated in the developing brain of preschoolers, when language abilities are rapidly emerging. The current study therefore examined whether phase-specific procedural learning predicts fine-grained grammar and vocabulary abilities across comprehension and production in Cantonese-speaking preschoolers, integrating Day 1 and Day 2 behavioral trajectories with task-based fMRI during Day 1. **Method.** Fifty-six Cantonese-speaking 4-year-olds ($M = 4.54$ years) completed a two-day Serial Reaction Time task (Thomas et al., 2004) with interleaved Sequence-Random blocks (8-item sequence; Day 1: six Sequence-Random block-pairs; Day 2: two Sequence-Random block-pairs). A PM index was computed at each block-pair as the z-scored Random–Sequence reaction time difference, with individual learning trajectories (slopes) extracted for Day 1 (initial acquisition) and Day 2 (post-learning re-engagement). Language outcomes included grammar comprehension, grammar production, vocabulary production from Hong Kong Test of Preschool Oral Language (Wong et al., 2019), plus receptive vocabulary from the PPVT (Dunn & Dunn, 2007). A mixed-effects model examined Day 1 and Day 2 slopes as predictors of language across modality (comprehension vs. production) and domain (grammar vs. vocabulary), controlling for age, nonverbal IQ, and gender. The final three Sequence-Random block-pairs on Day 1 were performed inside the MRI scanner (n = 19 after motion scrubbing). Preliminary whole-brain group-level analysis examined the Sequence-Random contrast across all block-pairs (voxel-wise $p < .02$, cluster-level FWE $\alpha < .05$, $k \geq 219$ voxels). **Results.** Behaviorally, the Day 1 trajectory predicted all four language outcomes ($b = 2.79$, $p = .01$), with no significant